Phase 1: Preprocessing Strategy Report

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1. Introduction

This project develops a classifier that assigns a fetal ultrasound image to one of four standard anatomical planes (abdomen, brain, femur or thorax), so that trained nurses and midwives in low-resource settings receive immediate feedback while they acquire images, with low-confidence predictions referred for human review.

The model will be pretrained on FETAL_PLANES_DB (the source dataset), which contains 12,400 images from two hospitals in Barcelona [3, 4], and then fine-tuned on the African maternal-fetal planes dataset (the target dataset), which contains 450 images from Algeria, Egypt, Ghana, Malawi and Uganda [1, 2]. The two datasets were acquired with different scanners and settings, so their image quality differs, and the preprocessing decisions below take this difference into account.

2. Dataset Audit

A preliminary audit of the dataset was conducted on the raw data to inform the planned strategies.

2.1 Dimensions

Table 1: Dimensions of the two datasets.

	Target dataset	Source dataset
Metadata matrix (N × P)	450 × 5	12,400 × 7
Patients	127	1,792
Sites	5 countries	2 hospitals, 4 machine types
Restricted to the four target planes	450 images	6,561 images, 1,158 patients
Image sizes	One per country, 400–600 px wide, aspect ratio 0.89–1.25	10 sizes, 640–1495 px wide, aspect ratio 1.44–2.17
Colour mode	350 grayscale, 100 RGB (all from Egypt)	All grayscale

Since Patient_num restarts at 1 in each country of the target dataset, I will identify patients by Center combined with Patient_num, which gives 127 patients in total as reported in the dataset publication. The audit also shows that each country contributed images of a single, unique size, so the image size alone identifies the country.

2.2 Variable Inventory

The table below shows an inventory of all variables in the target and source datasets and their roles. Center exists only in the target dataset, while US_Machine, Operator and Brain_plane are in the source dataset only.

Table 2: Variable inventory.

Column	Data type	Role	Justification for the role
Plane	Nominal	Target	The four planes have no natural order.
Patient_num	Identifier	Grouping key	Arithmetic on it is meaningless since it only keeps each patient's images together.
Center, US_Machine	Nominal	Stratification and audit only	Unavailable when a user uploads an image, and both would strengthen the acquisition shortcut.
Operator	Nominal	Audit only	Unavailable at inference, and 19.7% unknown.

Column	Data type	Role	Justification for the role
Brain_plane	Nominal	Dropped	Equal to "Not A Brain" for every non-brain image, so it reveals the label.
Train	Binary	Not used	The original authors' split, not a property of the image.
Filename, Image_name	Text identifier	File lookup only	Both contain the plane code and must never reach the model.
Image	Continuous	Model input	Pixel intensities from 0 to 255.

Neither dataset contains ordinal, discrete-count or datetime columns. Although pixel values are stored as integers, they are quantised measurements of a continuous echo-intensity signal, so I will treat them as continuous.

2.3 Missingness and Integrity

Both datasets have complete metadata as every column has 0 missing values (0.00%), so none of the 2,250 target cells or the 86,800 source cells is missing, and all 450 target rows and 12,400 source rows are complete. Every row matches exactly one readable image file, with no orphan files and no duplicate filenames, and the plane code embedded in each filename agrees with its label in all 12,850 rows. Because the labels are image-level, there are no bounding boxes or masks whose geometry would need verifying.

However, the preliminary audit still revealed some implicit forms of missingness:

1. The source dataset records unknown values as the category "Other", which affects Operator in 19.7% and US_Machine in 3.7% of the four-plane subset.
2. Ghana and Uganda contributed no thorax images in the target dataset.
3. The 100 Egyptian RGB images are effectively grayscale. In 99 of them, the three channels are identical, and the remaining one contains no coloured pixels, so converting them to grayscale will lose no information.

2.4 Target Variable

The task is multi-class classification with four mutually exclusive classes. I define the imbalance ratio as the size of the largest class divided by the size of the smallest.

Table 3: Class distribution.

Class	Target	Source (four planes)
Abdomen	125 (27.8%)	711 (10.8%)
Brain	125 (27.8%)	3,092 (47.1%)
Femur	125 (27.8%)	1,040 (15.9%)
Thorax	75 (16.7%)	1,718 (26.2%)
Imbalance ratio	1.67 : 1	4.35 : 1

The target imbalance is mild and extrinsic [6], since thorax is under-represented not because the plane is rare in practice but because two countries did not collect it. On the other hand, the source imbalance is moderate and dominated by the brain class.

Decision. I will train with a class-weighted cross-entropy loss whose weights are inversely proportional to class frequency, and I will not resample. I will compute the weights on each training fold only, so that no information from validation or test data enters training. I will not oversample because it would repeat the same few thorax scans, about 61 after de-duplication, and encourage memorisation. Also, I am not leaving the data uncorrected because the 4.35 : 1 source imbalance would bias pretraining towards the brain class. I will evaluate with macro-F1 as the primary metric, because it weights all four planes equally and a missed femur is as costly to the user as a missed brain, alongside per-class recall and calibration error, since the threshold for referring predictions to human review depends on calibrated confidence.

3. Cleaning Strategy

3.1 Duplicate Images

In the preliminary audit, I was able to detect duplicate copies of images by hashing every file (MD5), and near-copies by computing a 64-bit perceptual hash of each image, comparing pairs by the Hamming distance between their hashes, where 0 means visually identical. Table 4 summarises what the audit found.

Table 4: Duplicate detection results (four-plane subset for the source).

	Target	Source
Exact duplicate pairs or groups	47	113
Further near-duplicate pairs (distance ≤ 4)	7	466
Duplicates filed under different patients	54 (47 + 7)	30
Duplicates crossing the provided train/test split	25	1
Groups with conflicting plane labels	0	4

In the target dataset, 108 of the 450 images (24%) belong to a duplicate pair, which means 54 images (12%) are redundant copies. Every copy is attributed to a different patient, and 25 pairs cross the provided train/test split, so a model evaluated on that split would partly be tested on images it had already seen during training.

Decision.

1. **Threshold.** Exact hashing misses copies that were re-saved or slightly altered, so I will also treat two images as duplicates when their perceptual-hash distance is 4 or less. Visual inspection confirmed that each of the seven target pairs within this threshold is the same scan stored twice, whereas the first pair with different labels appears at distance 6, where the hash begins to match different scans.
2. **Handling.** I plan to remove duplicates and near-duplicates before splitting, keeping one copy of each duplicate group, so that no image appears in both the training and test splits, as recommended in the lecture on data leakage. I will record every exclusion in a log, because duplicates add no independent information and can inflate evaluation results.
3. **Conflicting labels.** The four conflicting source groups are an instance of label multiplicity, where the same image has more than one label. Without expert review to resolve them, I will drop them, because keeping either version risks training on incorrect ground truth.

I expect about 54 target images (12%) to be removed, about 14 of them thorax images, together with about 500 source images, a loss I accept because a credible evaluation matters more than nominal dataset size.

3.2 Missingness Mechanisms

Table 5: Missingness mechanisms and treatment.

Case	Mechanism	Evidence	Treatment
Operator recorded as "Other"	MAR	Follows the observed machine: every Voluson S10 and every unknown-machine image lacks an operator, and each named operator used one machine only.	No imputation; audit-only column.
US_Machine recorded as "Other"	MAR	226 of 240 such images (94%) have dimensions produced only by the Voluson S10, so observed image size explains the missingness.	No imputation; kept as "Unknown".
Brain_plane recorded as "Other"	Not missing	A valid category for a non-standard brain plane.	None; column dropped.

Case	Mechanism	Evidence	Treatment
No thorax images from Ghana or Uganda	MAR (site level)	Fully explained by the observed Center.	Handled through evaluation design.

None of the cases is MCAR, because each follows an observed variable, and none is MNAR, because no value is missing as a result of what that value would have been.

I will neither impute nor delete any of these values. Imputation is unnecessary because none of the affected columns is a model input, so mean, median, mode and KNN imputation do not apply, and row deletion is only appropriate when values are missing completely at random and affect very few rows; under MAR, deleting rows would bias the data against the machines and sites involved. The missing thorax images cannot be imputed at all, so I will account for them in the evaluation design:

- I will report thorax performance only for the three countries that contributed thorax images.
- I will stratify splits by plane rather than by plane and country, because the combined strata would contain empty cells.
- I will use leave-one-acquisition-domain-out evaluation to test whether the model has learned that images resembling Ghana or Uganda are never thorax.

3.3 Outliers and Image Quality

For images, I define an outlier as an image that is badly exposed, nearly blank or atypical for its acquisition setting, and I will measure this through each image's mean pixel intensity. In the audit this measure is right-skewed, with skewness 0.90 in the target and 0.57 in the source. When the IQR and Z-score rules are applied to the pooled target data, they flag 7 and 4 images respectively, and every flagged image is Egyptian. Visual inspection of the extremes shows clear Egyptian scans acquired at high gain and underexposed Ghanaian scans whose anatomy remains visible, with a single Ghanaian femur image that is barely legible.

Decision.

1. **Detection rule.** I will flag an image when its mean intensity falls outside $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$, with Q1, Q3 and IQR computed within each site on the training data. I prefer the IQR rule because it is robust to skew, whereas the Z-score rule assumes normality and is distorted by the very extremes it seeks to detect. Computing the bounds within each site makes the rule identify images that are unusual for their own acquisition setting, instead of flagging an entire country for being brighter.
2. **No deletion.** I will not delete flagged images, because none is corrupted or blank, and dark and bright scans are what the deployed tool is expected to encounter in the real-world environment.
3. **What a flag triggers.** In Phase 2, I will report per-class results separately for flagged images, to check whether exposure affects performance.
4. **Treatment.** I will clip each image's pixel intensities to that image's own 1st and 99th percentiles, a pixel-level form of winsorisation that caps overlay pixels and speckle without altering anatomy (Section 4.2).

3.4 Association, Redundancy and Confounding

With about 50,000 pixels per image and neighbouring pixels correlated by construction, pixel-level correlation and VIF are not meaningful for this model input. For image data, the redundancy that matters is duplication, which Section 3.1 addresses, and the confounding that matters is whether the label can be predicted from signals unrelated to anatomy. Figure 1 shows that many images carry such signals: Malawi images include a logo, Algerian images include corner text, Ghanaian images show caliper marks, and Ugandan images show calipers together with measurement labels. This acquisition signature likely reflects the equipment and settings used rather than the location: within the Spanish dataset alone, the machine that produced an image can be identified 99% of the time (Table 6). It is a form of leakage from the data generation process [5]. Huyen describes COVID-19 models that learned hospital-specific text fonts instead of pathology, which is the same risk posed by the logos and calipers in these images.

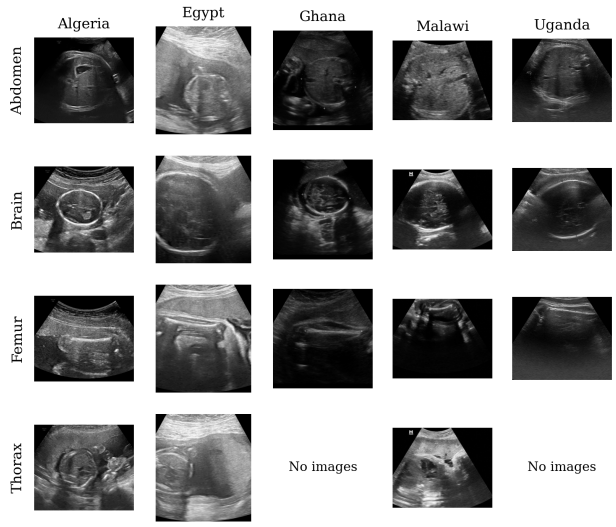


Figure 1: One image per plane and country from the target dataset.

To quantify this risk, the audit shrank every image to 32 × 32 pixels, which removes the size difference between countries, and trained a logistic regression with patient-grouped five-fold cross-validation after duplicates were excluded. Table 6 shows that the country can be predicted far more accurately than the plane, so a convolutional network could exploit the same acquisition signature instead of learning anatomy. I regard this as the most important generalisation risk identified in the audit.

Table 6: Shortcut test on 32 × 32 thumbnails.

Dataset	Predicted variable	Zero-rule baseline	Accuracy
Target	Plane	28.5%	56.2%
Target	Country	23.0%	93.4%
Source	Plane	49.0%	72.5%
Source	Machine	68.0%	99.2%

Planned association tests. In Phase 2 I will run the tests listed in Table 7 against Plane. Because none of the metadata columns is a model input, these tests will quantify confounding, that is, how strongly site or operator is entangled with the label, rather than select features.

Table 7: Planned feature-target tests.

Variable	Type	Test against Plane	Purpose
Center, US_Machine, Operator	Nominal	Chi-square test, Cramér's V	Quantify confounding between site and label
Image summary features	Continuous	Mutual information	Rank features by association with the label

Chi-square and Cramér's V are used for pairs of nominal variables. For continuous features against the nominal Plane, I chose mutual information over ANOVA F, because ANOVA assumes normally distributed groups and these features are skewed. Pearson and point-biserial correlation do not apply, because Plane has four unordered classes.

Multicollinearity protocol. For diagnostics, I will compute eight summary features for each image: mean intensity, standard deviation of intensity, entropy, sharpness, fan area fraction, width, height and aspect ratio. I will treat a pair of

features as collinear when the absolute Spearman correlation between them is 0.85 or more, using Spearman because several features are skewed, and I will also flag any feature whose variance inflation factor exceeds 10. When two features are collinear, I will keep the one with the higher mutual information with Plane, then recompute VIF and repeat until no feature exceeds the threshold. I expect width, height and aspect ratio to be collinear, since all three are determined by site. These features will describe the data rather than feed the model, and they will also let me compare image quality between the two datasets. I will run principal component analysis on the backbone's image embeddings, standardised with statistics from the training fold, rather than on raw pixels.

Decision. I will reduce and monitor the acquisition shortcut with three complementary measures:

1. **Mask outside the fan.** I will set every pixel outside the ultrasound fan to black, locating the fan as the largest bright connected region of each image, which removes logos and scanner text without touching anatomy.
2. **Augment the training data.** I will apply photometric and mild geometric augmentation to training batches only, including quality degradation such as blur, noise, and contrast and gain changes, so that equipment-specific appearance stops being a reliable cue and the source images cover the range of image quality seen in the target data. I will fix the exact transformations in Phase 2.
3. **Measure the remaining risk.** I will report leave-one-acquisition-domain-out results, repeat the shortcut test after preprocessing, and check that adding or removing logos and caliper marks does not change the predicted plane.

Caliper marks inside the fan cannot be removed without damaging anatomy. I record this as a limitation, noting that deployed images are assessed before measurements are taken and will therefore rarely contain calipers.

4. Encoding, Standardisation, Split and Leakage Plan

4.1 Label Encoding

I will encode Plane as integers (abdomen 0, brain 1, femur 2, thorax 3), an alphabetical order that matches the plane codes already used in the target filenames. These integers are class identifiers only, and the cross-entropy loss treats them as unordered categories. No ordinal or one-hot encoding is needed, because no categorical column is a model input. Had the low-cardinality site columns been inputs (5 countries, 4 machines), one-hot encoding would have suited them; label encoding applies only to the target.

4.2 Image Standardisation

Table 8 lists the standardisation steps, which I plan to apply in order to every image. Steps 1, 3 and 5 make all images share one format and resolution, which, as Huyen notes, makes it harder for models to know which image is from which scan machine [5]. I will not apply a log transform, because scanners already log-compress B-mode images and a second logarithm would only distort contrast. I plan to match quality by degrading the source images during training, instead of upscaling the low-quality images. I rejected standardising with training-set statistics because it would shift inputs away from the scale that pretrained backbones expect. I also rejected per-image standardisation, because the audit showed that it barely changes how well the country can be predicted from an image (93.4% to 90.7%), so I rely on brightness augmentation instead to stop exposure acting as an acquisition cue. Since every step is either per-image or uses fixed constants, preprocessing involves no fitting and cannot leak information between splits.

Table 8: Image standardisation steps, applied in order.

Step	Operation	Justification
1	Convert to single-channel grayscale	Loses no information (Section 2.3) and removes the RGB cue unique to Egypt.
2	Mask outside the ultrasound fan	Removes logos and scanner text (Section 3.4).
3	Crop to the fan's bounding box	Removes background borders, so that later padding depends on fan shape rather than on the original image size, which identifies the country.

Step	Operation	Justification
4	Clip to the 1st–99th percentile, computed inside the fan only	Caps extreme pixels (Section 3.3); excluding the background prevents it from dragging the lower percentile to zero. Values are not stretched back to 0–255, because stretching would amplify speckle in underexposed images and erase genuine exposure differences.
5	Resize to 224 × 224, preserving aspect ratio and padding with black	Aspect ratios range from 0.89 to 2.17, and stretching would distort anatomy differently at each site.
6	Min-max scale to [0, 1] by dividing by 255	Pixel values have fixed known bounds, so no fitting is required.
7	Replicate to three channels and standardise with the selected backbone's pretraining mean and standard deviation	The candidate backbones (ResNet18, MobileNetV2 and a YOLO classification model) are pretrained on ImageNet, and each uses its own fixed pretraining constants.

4.3 Split Strategy

Table 9: Planned partitions.

Dataset	Partition	Share	Purpose
Target	Development	80%	Five-fold cross-validation for tuning
Target	Held-out test	20%	Final evaluation, used once (about 12 thorax images expected)
Source	Training	85%	Pretraining
Source	Validation	15%	Early stopping and checkpoint selection

Ratios. With about 400 target images remaining after de-duplication, a fixed 15% validation set would hold about 60 images and only nine thorax images, which is too few for a stable estimate. I will therefore run five-fold cross-validation inside the development set, so that every development image is used for validation once. The source dataset serves only for pretraining, so it needs no test set.

Stratification and grouping. I will split both datasets with StratifiedGroupKFold, stratifying by plane and grouping by patient, so that no patient appears in more than one partition. Because plane-by-country strata would contain empty cells, I will instead check the country balance of the test set after splitting.

Temporal ordering. Neither dataset contains acquisition dates, so a temporal split is neither possible nor required.

Validation. I will tune hyperparameters with five-fold StratifiedGroupKFold on the development set. As a secondary evaluation, I will hold out one acquisition domain at a time, training on the others and testing on the held-out one, which measures generalisation to unseen equipment directly. Because the target data does not record the machine used, I will use country as the proxy for acquisition domain, since all images from a country share a single format.

Provided split. I will not use the provided Train column, for two reasons:

- **Target dataset:** it assigns only 217 of the 450 images (48%) to training, and 25 duplicate pairs cross it.
- **Source dataset:** it places 98% of Voluson S10 images in its test partition, so a model trained on it would barely see that machine.

4.4 Leakage Prevention Plan

The audit identified three leakage risks: duplicate images filed under different patients (Section 3.1), multiple images from the same patient (Section 4.3), and the acquisition signature that lets the site be predicted from the image (Section 3.4). I will prevent leakage between partitions in eight ways:

1. De-duplication will run before splitting and will fit no statistics.

2. I will check for duplicates again after splitting, to confirm that no image appears in more than one partition.
3. I will hash across the source and target datasets and drop the source copy of any match, so that no target image appears in pretraining.
4. The split will be made once, by patient, with a fixed seed, and each image's partition will be saved for every later step.
5. Preprocessing will be fit-free (Section 4.2).
6. Class weights and model weights will be fitted on training folds only.
7. The confidence threshold for human review will be fitted on out-of-fold validation predictions.
8. Augmentation will apply only to training batches.

The target test set will never be used for pretraining, fine-tuning, tuning or threshold selection.

5. Planned Pipeline

1. Load the metadata, build patient keys, and restrict the source dataset to the four target planes.
2. Remove duplicates (perceptual-hash distance ≤ 4) within and across datasets, dropping the source copy of any cross-dataset match so that the target data stays intact, drop conflicting-label groups, and re-check for duplicates after splitting.
3. Encode labels as 0–3.
4. Split by patient, stratified by plane, and save each image's partition.
5. Standardise each image: grayscale, fan mask, crop to the fan, percentile clipping within the fan, padded resize to 224×224, [0, 1] scaling and backbone standardisation.
6. Augment training batches only, and compute class weights on each training fold.
7. Tune with five-fold grouped cross-validation, and evaluate leave-one-acquisition-domain-out together with the logo and caliper invariance check.
8. Evaluate once on the target test set with macro-F1, per-class recall and calibration error.

References

[1] C. Sendra-Balcells et al., "Generalisability of fetal ultrasound deep learning models to low-resource imaging settings in five African countries," *Scientific Reports*, vol. 13, p. 2728, 2023.

[2] C. Sendra-Balcells et al., "Maternal fetal ultrasound planes from low-resource imaging settings in five African countries," Zenodo, 2023. doi:10.5281/zenodo.7540447.

[3] X. P. Burgos-Artizzu et al., "Evaluation of deep convolutional neural networks for automatic classification of common maternal fetal ultrasound planes," *Scientific Reports*, vol. 10, p. 10200, 2020.

[4] X. P. Burgos-Artizzu et al., "FETAL_PLANES_DB: Common maternal-fetal ultrasound images," Zenodo, 2020. doi:10.5281/zenodo.3904279.

[5] C. Huyen, *Designing Machine Learning Systems*. Sebastopol, CA: O'Reilly Media, 2022.

[6] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Transactions on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263–1284, 2009.